THE STRUCTURE OF MATHEMATICAL REALITY

Phase IV — Chapter 11

Probability, Universality, and Levels of Description

Exercises

Self-Study Curriculum

1. You bootstrap a statistic and its sampling distribution comes out approximately Gaussian, even though the raw data are visibly skewed. Explain why, using the CLT. Then describe a data situation from your field where the bootstrap distribution would *not* concentrate, where averaging is the wrong move, and say which condition (dependence or heavy tails) is responsible.

2. Temperature is a single sharp number, yet it summarizes 10^{23} molecules with a huge range of individual speeds. In terms of concentration of measure, why is the aggregate sharp while the parts are not? What does this tell you about when a "summary statistic" is trustworthy?

3. Pick a Bayesian and a frequentist statement you have actually made or read (e.g., a credible interval vs. a confidence interval). State precisely what each one claims and, in terms of the three interpretations, why they are *not* the same claim even when the numbers coincide.

4. Universality says the microscopic details often do not matter for the large-scale law. Give one example from your own work where this is a *gift* (a simple model works despite ignoring mechanism) and one where it is a *trap* (you were fooled into thinking you understood the mechanism because a coarse model fit).